



Virtual Copy



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Introduction

- ▶ An array-based solution, HPE 3PAR StoreServ Storage with HPE 3PAR Virtual Copy Software eliminates the cost and performance overhead associated with host-based snapshots.
- ▶ The unique performance advantages of HPE 3PAR StoreServ Storage ensure that users can leverage PIT snapshots without the performance concerns that impact traditional array-based copies.



Virtual Copy

- ▶ HPE 3PAR Virtual Copy Snapshots are also highly flexible.
- ▶ Users can make hundreds of snapshots per base volume, enabling histories of application data to be archived online efficiently.
- ▶ Users also have the option to create read-only (RO) or read-write (RW) snapshots of snapshots.
- ▶ Therefore, organizations can selectively test, use, and change production-quality data safely and easily while protecting production data and other snapshots from accidental or intentional changes.



Virtual Copy

- ▶ Administrative overhead is dramatically reduced and organizational productivity is greatly improved.
- ▶ The software features a “promote” capability that allows users to take any snapshot associated with a base volume and “promote” it to a base volume, simply and quickly.
- ▶ In this way, a base volume can be rapidly synchronized to any one of its associated snapshots, effectively “rolling back” the base volume to a previous point in time.



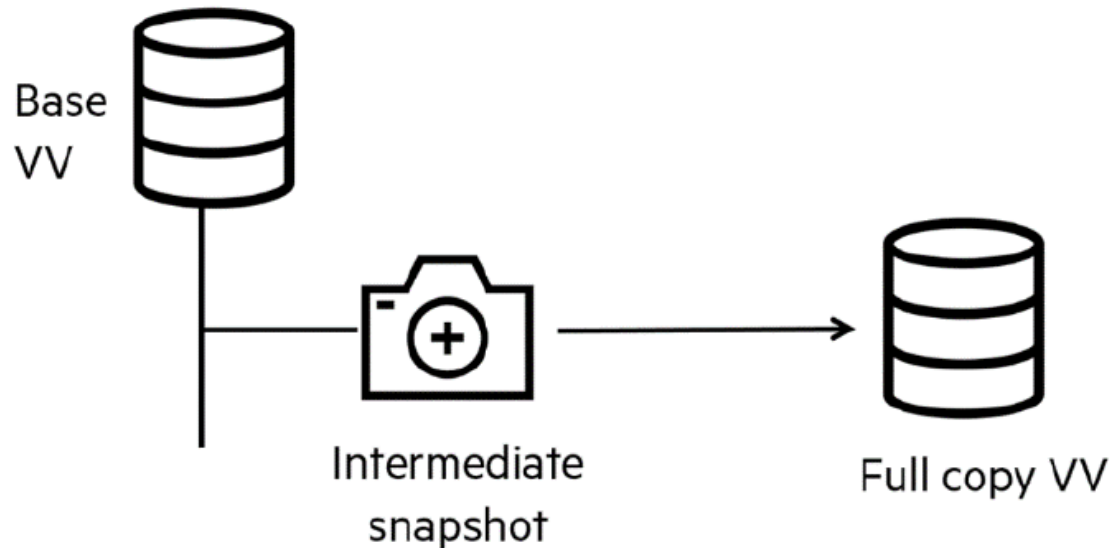
HPE 3PAR Virtual Copy basics

- ▶ HPE 3PAR StoreServ Storage supports two technologies to provide users with copies of HPE 3PAR virtual volumes (base VVs):
 - ▶ Physical copies (clones)
 - ▶ Virtual Copy Snapshots
- ▶ Each of these technologies allow users to access a PIT copy of a base VV and to read data from or write data to that PIT copy.



Physical copies

- ▶ A physical copy is a full copy of a volume. A physical copy duplicates all the data from one original base volume to a destination volume.
- ▶ Any change to either volume causes them to lose synchronization with each other, which can be corrected by resynchronizing the two volumes.



Physical Copies

- ▶ Physical copies can be online or offline:
- ▶ With online physical copies, the destination volume is automatically created and can be exported immediately to the host as cloning happens in the background.
- ▶ Offline physical copies require a destination volume with a user space size at least as large as the user space of the base volume being copied.
- ▶ Offline physical copies cannot be exported until the data copying is complete.



Virtual Copy Software

- ▶ HPE 3PAR Virtual Copy Software is an HPE 3PAR snapshot implementation used to provide a PIT virtual copy of a VV.
- ▶ It also ensures that the original data can always be obtained if a problem occurs when updating the data on a VV.
- ▶ HPE 3PAR Virtual Copy implements an efficient variant of copy-on-write (COW) and redirect-on-write (ROW) methodologies.



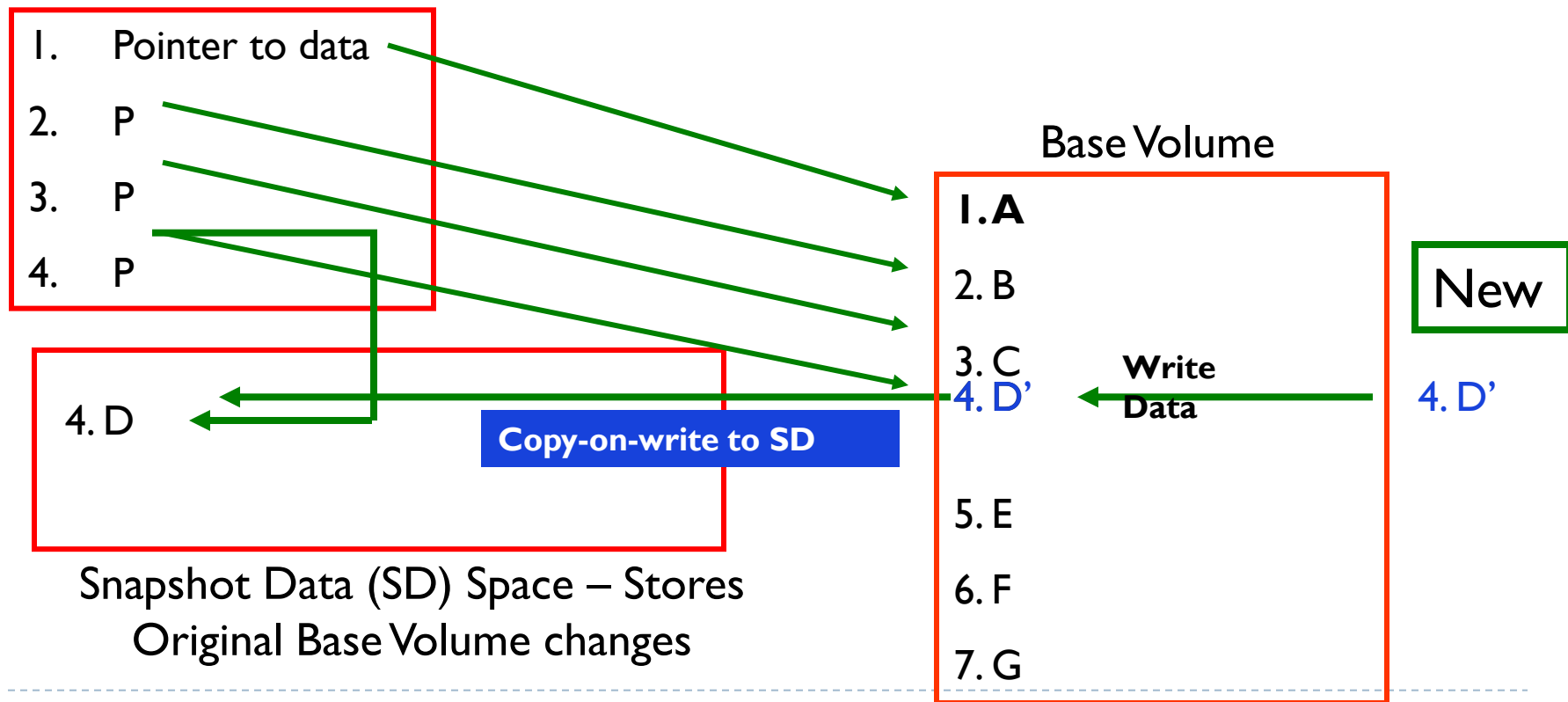
Virtual Copy Software

- ▶ HPE 3PAR Virtual Copies are always thin, non-duplicative (only one copy of the change blocks is kept and then referenced).
- ▶ A Virtual Copy can have variable quality of service (QoS) because the user can specify a different CPG for snapshot data, such as user data (SD) on an SSD and snapshot data (SD2) on a more cost-efficient SAS tier.
- ▶ Snapshots of TDVVs that reside on an SSD tier deduplicate snap data.



Virtual Copy – Copy-on-Write Function (Data Written)

Snapshot Admin (SA) Space Timestamp 11/25/11 14:35

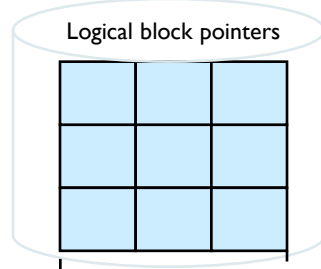


Virtual Copy Snapshot (Read Only)

snapshot create

Base Virtual Volume (VV)

Read/Write (RW)

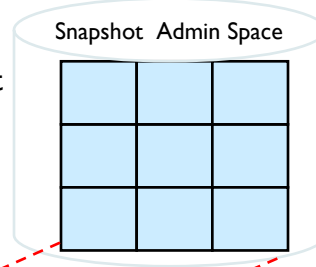


Create
Read Only Snapshot



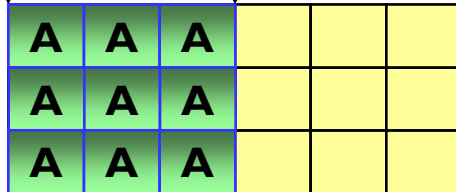
Snapshot (S0)

Read Only (RO)

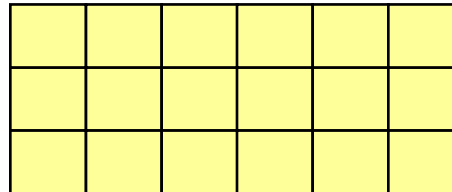


[Create Read Only Snapshot]

- Create the pointer at Snapshot Admin Space (AS)
- All pointers point to Base Virtual Volume chunklet



Base Virtual Volume CPG



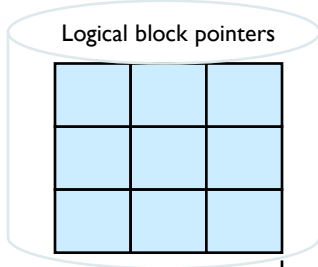
Snapshot CPG

Virtual Copy Snapshot (Read Only)

copy on write (update Base VV)

Base Virtual Volume (VV)

Read/Write (RW)

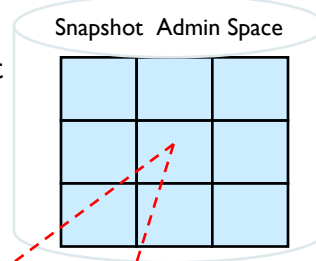


Create
Read Only Snapshot



Snapshot (S0)

Read Only (RO)

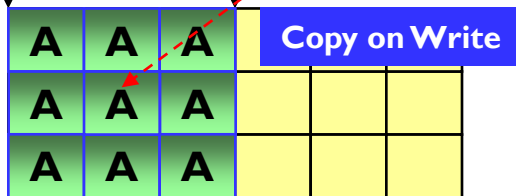


[Update base vv after creating read only snapshot]

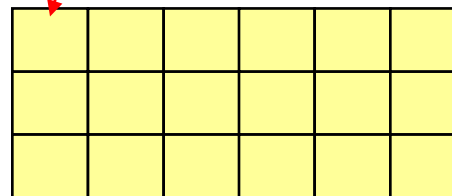
- Copy old data to Snapshot Data Space(SD)
(Copy on Write)
- Change the SA pointer from Base vv to SD
- Update base vv

B

New Data

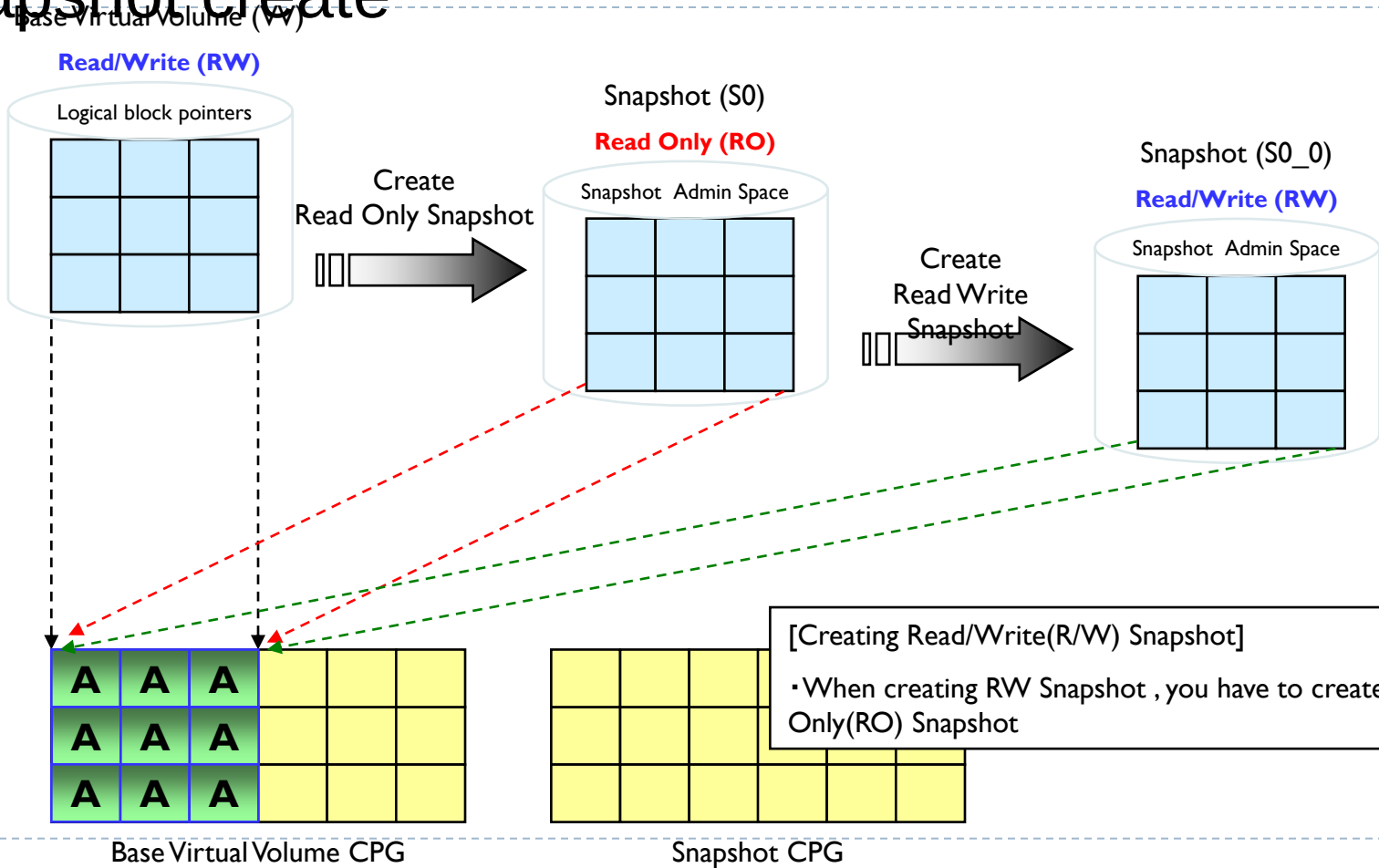


Base Virtual Volume CPG



Snapshot CPG

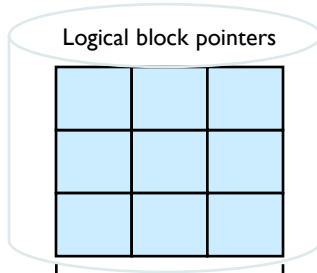
Virtual Copy Snapshot (Read Write) snapshot create



Virtual Copy Snapshot (Read Write) copy on write (update Base VV)

Base Virtual Volume (VV)

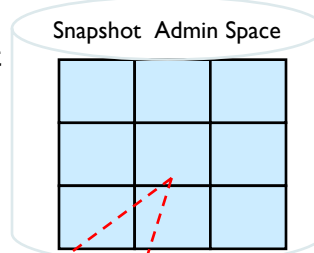
Read/Write (RW)



Create
Read Only Snapshot

Snapshot (S0)

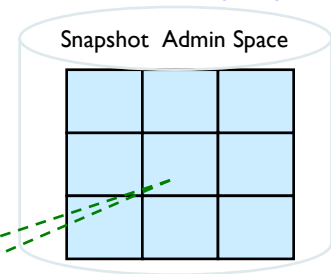
Read Only (RO)



Create
Read Write
Snapshot

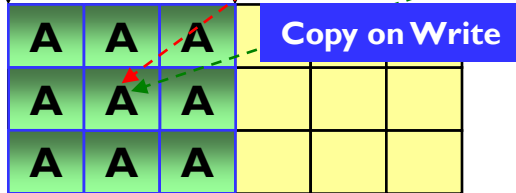
Snapshot (S0_0)

Read/Write (RW)

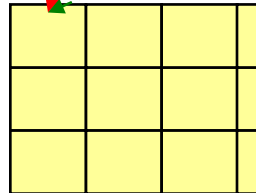


B

New Data



Base Virtual Volume CPG



Snapshot CPG

[Update Base vv after creating Snapshot]

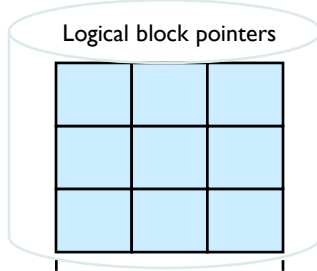
- Copy old data to Snapshot Data Space(SD)
(Copy on Write)
- Change the SA pointer from Base vv to SD
- Update base vv

Virtual Copy Snapshot (Read Write)

copy on write (update Snapshot S0_0) -1

Base Virtual Volume (VV)

Read/Write (RW)

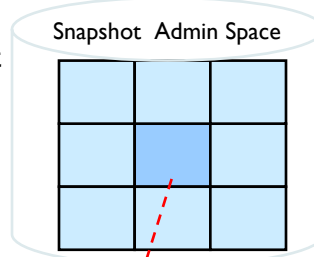


Create
Read Only Snapshot



Snapshot (S0)

Read Only (RO)

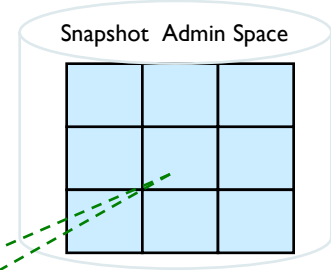


Create
Read Write
Snapshot

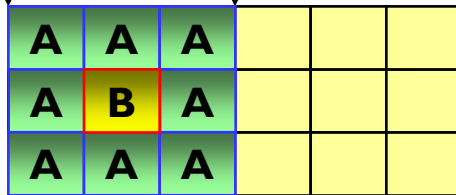


Snapshot (S0_0)

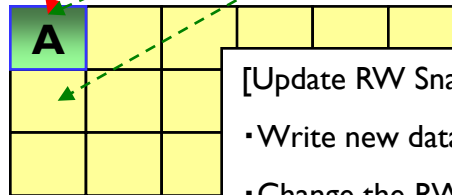
Read/Write (RW)



New Data



Base Virtual Volume CPG



Snapshot CPG

[Update RW Snapshot after creating snapshot -1]

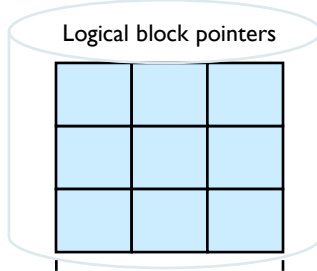
- Write new data to Snapshot Data Space(SD)
- Change the RW snapshot SA pointer to SD

Virtual Copy Snapshot (Read Write)

copy on write (update Snapshot S0_0) -2

Base Virtual Volume (VV)

Read/Write (RW)

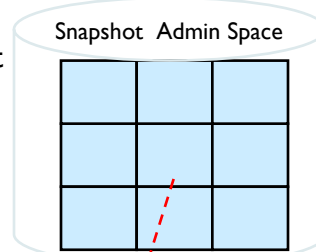


Create
Read Only Snapshot



Snapshot (S0)

Read Only (RO)

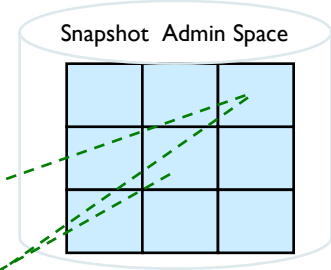


Create
Read Write
Snapshot



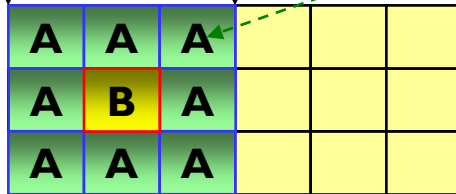
Snapshot (S0_0)

Read/Write (RW)

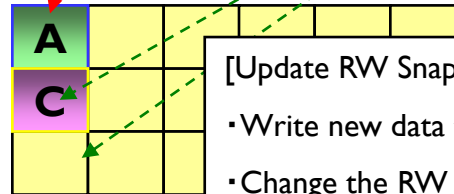


D

New Data



Base Virtual Volume CPG

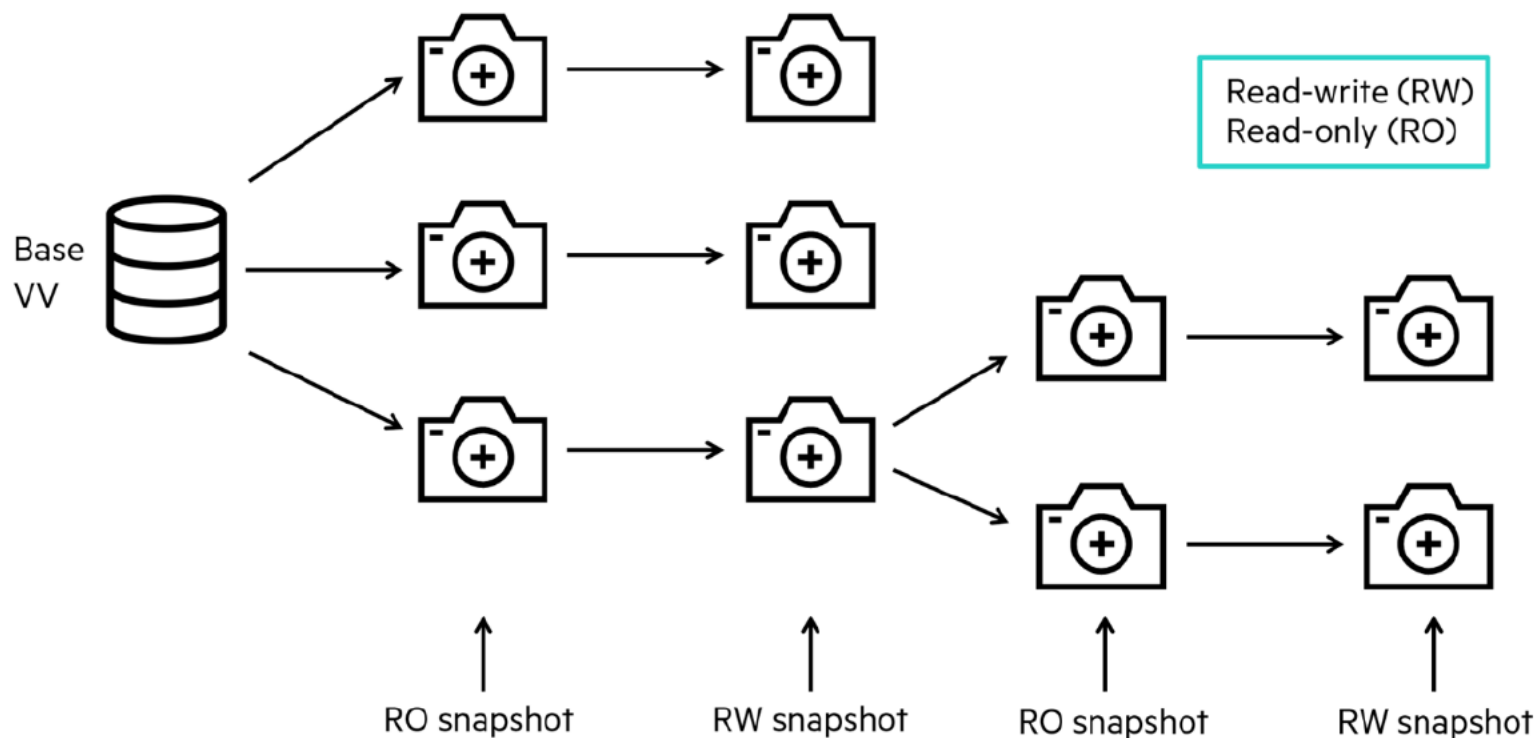


Snapshot CPG

[Update RW Snapshot after creating snapshot -2]

- Write new data to Snapshot Data Space(SD)
- Change the RW snapshot SA pointer to SD

Virtual Copies of Snapshot



Virtual Copy Max

The maximum values as of the HPE 3PAR OS 3.3.1 release are listed in Table 1.

Table 1. Maximum snapshots per VV

Maximum	Value
Maximum snapshot per VV: RO	2,048
Maximum snapshot per VV: RW	2,048
Maximum RW snapshot per RO snapshot	2,048
Maximum VVs per consistency group for group snapshot	300



Recovery

- ▶ With Virtual Copy Software, users can make frequent online copies of production data sets.
- ▶ If an inconsistency or deletion is detected, an administrator can restore from a “clean” copy made just before the inconsistency or deletion occurred.
- ▶ With this safety mechanism, users can minimize the costs of downtime and improve service levels for business users both economically and with reduced opportunity for error.



Recovery

- ▶ Depending on the application requirements, recovery can be accomplished in one of two ways:
- ▶ By promoting a “clean” snapshot to the base volume
- ▶ By immediately mounting and returning to production an instant RVW snapshot (taken from the same “clean” snapshot).
- ▶ This can be followed by a later promotion of the RVW snapshot to the original base volume when downtime is convenient.



Scheduling

- ▶ Virtual Copy Snapshots can also be regularly scheduled so that PIT copies of user data can be created and deleted over time.
- ▶ The running of periodic scheduled tasks to create or delete snaps enables flexibility in managing user data and can automatically remove volumes that have expired.
- ▶ Periodically removing snapshots enables customers to create and remove snapshots at regular intervals.
- ▶ The HPE 3PAR Scheduler enhances management automation, reduces administration time, and decreases the chance of administrative error by giving users full control over automated Virtual Copy Snapshot creation and deletion.

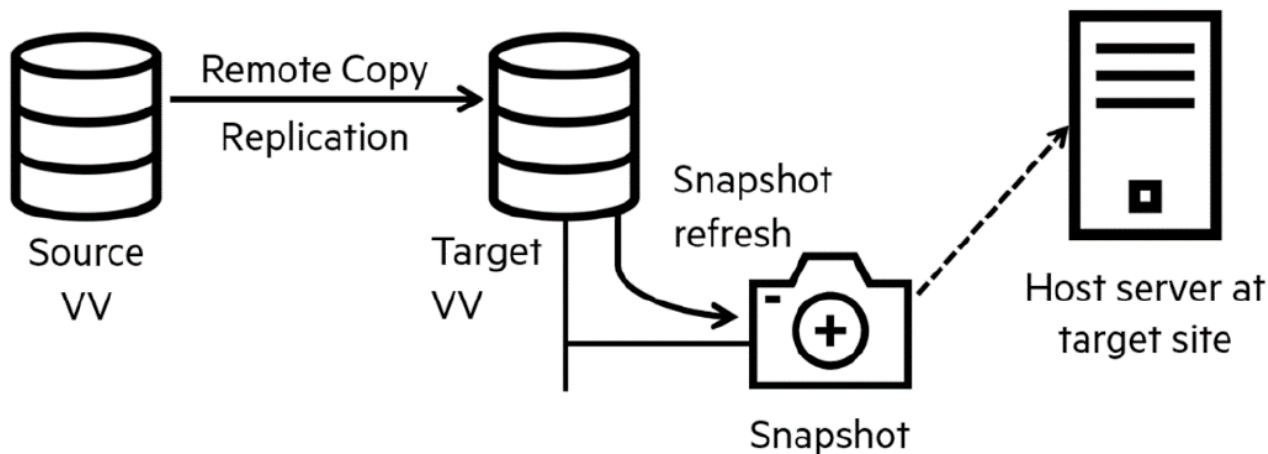


Refresh

The HPE 3PAR refresh feature can also be used to host an application that requires a copy of production data at a target site of a disaster recovery solution.

In an HPE 3PAR Remote Copy configuration, target volumes synchronize data source volumes. Because target volumes are write-protected, an RVV snapshot can be created and configured in the hosts at target site.

The snapshot can be refreshed at regular intervals by using the HPE 3PAR refresh feature to allow applications to get access to updated data.

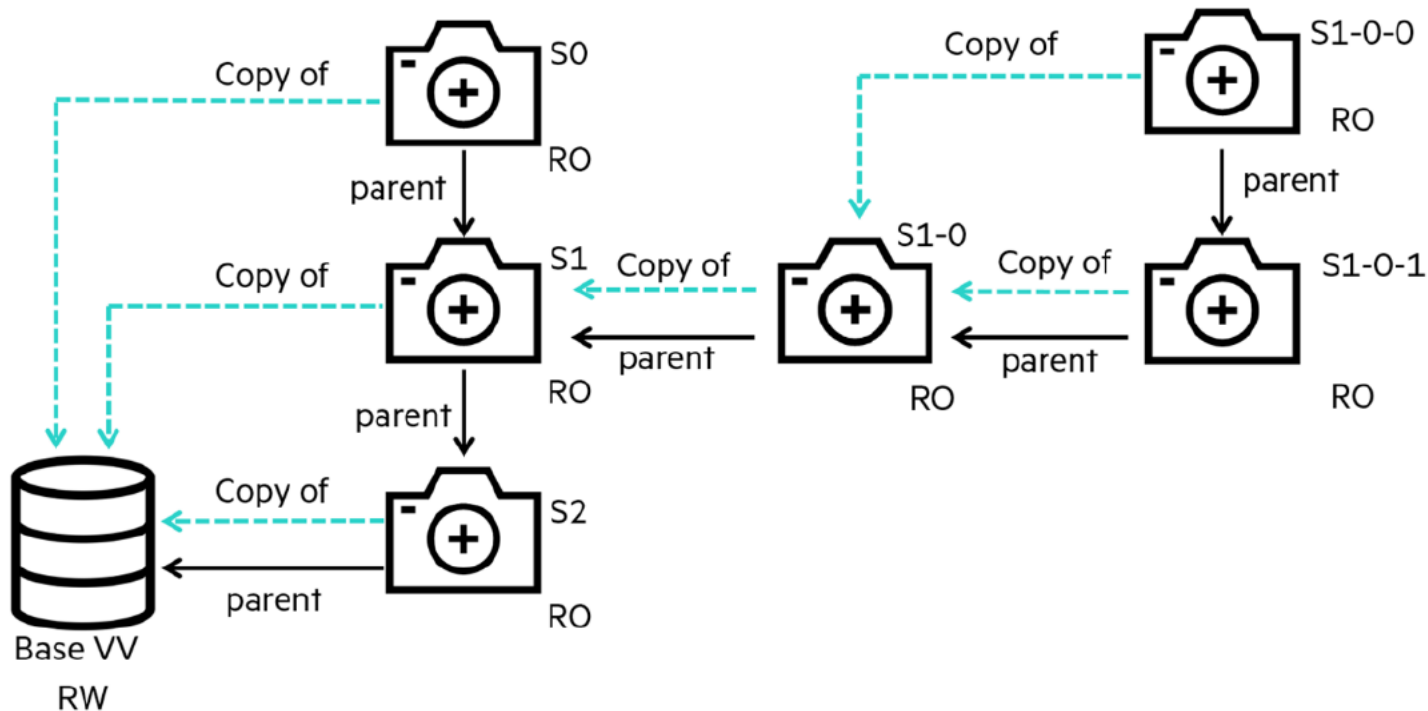


-
- ▶ A snapshot volume consists of SA and SD areas:
 - ▶ SA is the management (metadata) space that contains first level/second level (L1/L2) bitmaps. It indicates which SA and SD pages are used with the exception or allocation tables (L1/L2/L3) pointing to the snapshot location within the SD.
 - ▶ SD contains the actual user data. For a base VV, this is the “old” base volume data that is copied to the SD before the new data is written to the base volume; for a FPVV, TPVV, and TDVV, the SD contains both new and old data.



Virtual Copy Relationship

A virtual volume cannot be deleted if a child copy of it exists. For example, S1 cannot be removed unless S1_0, S1_0_0, and S1_0_1 are deleted first.



Creating consistent group snapshots of virtual volumes

- ▶ This operation creates consistent group snapshots of a list of VVs.
- ▶ Consistent group snapshots are all created at the same point in time so that related structures on different volumes in the group remain consistent.
- ▶ Below example, HPE 3PAR SSMC snapshots are created for the three VVs in a volume set.



Promoting a snapshot to a base volume

- ▶ This task copies the differences of a snapshot back to its base volume, allowing the user to revert the base volume to an earlier point in time.
- ▶ If the parent volume is exported, then the use of the `promotesv` command will fail.

```
3par_20850_3_185 cli% promotesv  
ha.3.rw Error: RW parent (ha.3)  
has exports defined
```

```
3par_20850_3_185 cli% promotesv ha.5.160131200257  
Task 3879 has been started to promote virtual copy ha.5.160131200257
```



Refreshing Virtual Copy

```
cli% showvv -s test*
```

						-----Snp-----					-----Usr-----				
						-- (MiB) --		- (% VSize) --			-- (MiB) --		- (% VSize) --		
Id	Name	Prov	Compr	Dedup	Type	Rsvd	Used	Used	Wrn	Lim	Rsvd	Used	Used	Wrn	Lim
65231	testvv	tpvv	No	No	base	1536	780	7.6	0	0	3584	2659	26.0	0	0
65241	testsv	snp	NA	NA	vcopy	--	*0	*0.0	0	0	--	--	--	--	--
2 total						1536	780				3584	2659			

```
cli%
```

```
cli% updatevv testsv
```

```
Updating VVs testsv
```

```
select q=quit y=yes n=no: y
```

```
cli% showvv -s test*
```

						-----Snp-----					-----Usr-----				
						-- (MiB) --		- (% VSize) --			-- (MiB) --		- (% VSize) --		
Id	Name	Prov	Compr	Dedup	Type	Rsvd	Used	Used	Wrn	Lim	Rsvd	Used	Used	Wrn	Lim
65231	testvv	tpvv	No	No	base	1536	0	0.0	0	0	3584	2659	26.0	0	0
65241	testsv	snp	NA	NA	vcopy	--	*0	*0.0	0	0	--	--	--	--	--
2 total						1536	0				3584	2659			

```
cli%
```



Refresh

```
cli% updatevv testsv
Updating VVs testsv
select q=quit y=yes n=no: y
select y=yes n=no: y
error: VV testvv has dedup accounting in process.
Command failed
```

updatevv on any vvol is currently failing with the above error.
cli%

The SSMC GUI interface does not provide options to perform a snapshot refresh.



Snapshot Policies

Create Policy General ?

General

Policy type ☐ Virtual volume ☒ Snapshot

Snapshot policy name

Snapshot name pattern +

Snapshot name VolumeName.170605141808

Snapshot schedules 1 Schedule

Schedule Name	Details	Retention Time	Expiration Time
Sched_1	0 minutes after the hour	—	— x

[Add schedule](#)

Additional Settings

▼ Snapshot

Access permissions

Volume ID Auto ID

Comments

Changed: Snapshot policy name to "Snap_pol_1" [Create](#) [Create +](#) [Cancel](#)

Create Snapshot Schedule ?

Schedule name

Schedule pattern

Starting at minutes after the hour

Alert ☒ Enabled

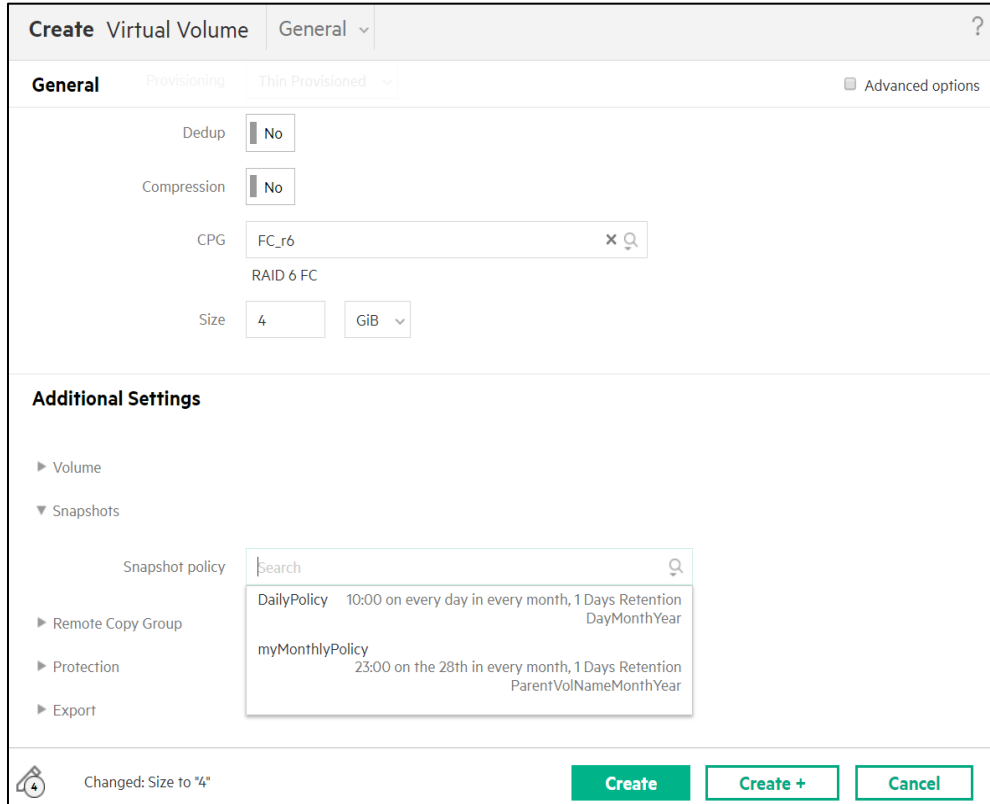
Retention time

Expiration time

Changed: Schedule name to "Sched2" [Add](#) [Add +](#) [Cancel](#)

- ▶ Snapshot policies are **NOT** array specific and can be reused on any array
- ▶ Snapshot policies are defined with a name pattern and schedules based on the desired frequency
- ▶ Choose Access Permission
- ▶ Create policy and assign to volumes and volume sets

Assign Snapshot Policy to Volume or Volume Set



Create Virtual Volume General ?

General Provisioning Thin Provisioned Advanced options

Dedup No

Compression No

CPG FC_r6 x

RAID 6 FC

Size 4 GiB

Additional Settings

Volume

▼ Snapshots

Snapshot policy search

DailyPolicy 10:00 on every day in every month, 1 Days Retention DayMonthYear

myMonthlyPolicy 23:00 on the 28th in every month, 1 Days Retention ParentVolNameMonthYear

Remote Copy Group

Protection

Export

Changed: Size to "4"

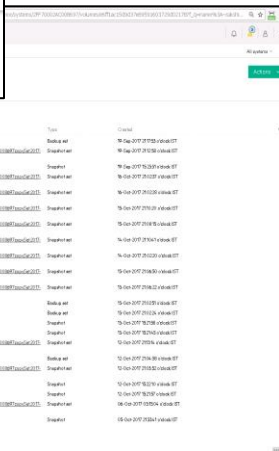
Create Create + Cancel

- ▶ Choose from an existing policy.
- ▶ Policy can be applied to single volume or a group of volumes
- ▶ If you choose to add the volume to a volume set, the policy will be applied to the entire volume set



Working with Restore Points





- ▶ **Restore Point** view shows all Restore Point for the selected Virtual Volume
- ▶ Link will point to “Restore Points” page highlighting specified object

Restore Points Overview

The screenshot displays the 3PAR StoreServ Restore Points Overview interface. The left pane shows a list of restore points with columns for Name, Type, System, and Creation Time. The right pane shows detailed information for the selected restore point, including General, Health, Policy, and Volumes sections.

Restore Points List (Left Pane):

Name	Type	System	Creation Time
ps2-36-backup-2017-10-1...	Backup set	---	15-Oct-2017 16:11:35 v/d...
SSMCRestoreVSet0002...	Backup set	---	30-Sep-2017 17:02:55 v/d...
SSMCRestoreVSet0002...	Backup set	---	20-Sep-2017 08:08:13 v/d...
SSMCRestoreVSet0002...	Backup set	---	12-Oct-2017 05:14:23 v/d...
SSMCRestoreVSet0002...	Backup set	---	26-Sep-2017 23:02:25 v/d...
SSMCRestoreVSet0002...	Backup set	---	30-Sep-2017 03:03:23 v/d...
SSMCRestoreVSet0002...	Backup set	---	27-Sep-2017 13:13:40 v/d...
SSMCRestoreVSet0002...	Backup set	---	02-Oct-2017 19:12:07 v/d...
Backup:2017-08-19 00:05:08	Backup set	---	19-Aug-2017 00:20:04 v/d...
SSMCRestoreVSet0002...	Backup set	---	03-Oct-2017 21:02:57 v/d...
SSMCRestoreVSet0002...	Backup set	---	22-Sep-2017 12:49:28 v/d...
SSMCRestoreVSet0002...	Backup set	---	01-Oct-2017 21:14:41 v/d...
SSMCRestoreVSet0002...	Backup set	---	24-Sep-2017 17:11:21 v/d...
SSMCRestoreVSet0002...	Backup set	---	02-Oct-2017 20:07:25 v/d...
SSMCRestoreVSet0002...	Backup set	---	18-Sep-2017 05:15:22 v/d...
SSMCRestoreVSet0002...	Backup set	---	03-Oct-2017 21:16:01 v/d...
SSMCRestoreVSet0002...	Backup set	---	30-Sep-2017 22:03:05 v/d...
SSMCRestoreVSet0002...	Backup set	---	23-Sep-2017 03:03:57 v/d...
SSMCRestoreVSet0002...	Backup set	---	29-Sep-2017 20:04:00 v/d...
SSMCRestoreVSet0002...	Backup set	---	01-Oct-2017 08:16:42 v/d...
SSMCRestoreVSet0002...	Backup set	---	01-Oct-2017 23:15:32 v/d...
SSMCRestoreVSet0002...	Backup set	---	02-Oct-2017 00:17:16 v/d...
SSMCRestoreVSet0002...	Backup set	---	25-Sep-2017 04:01:12 v/d...
SSMCRestoreVSet0002...	Backup set	---	09-Oct-2017 22:09:03 v/d...
SSMCRestoreVSet0002...	Backup set	---	07-Oct-2017 16:04:20 v/d...
SSMCRestoreVSet0002...	Backup set	---	21-Sep-2017 13:03:50 v/d...
SSMCRestoreVSet0002...	Backup set	---	27-Sep-2017 08:06:47 v/d...
SSMCRestoreVSet0002...	Backup set	---	27-Sep-2017 15:07:44 v/d...
SSMCRestoreVSet0002...	Backup set	---	28-Sep-2017 02:01:43 v/d...
SSMCRestoreVSet0002...	Backup set	---	06-Oct-2017 23:14:28 v/d...
SSMCRestoreVSet0002...	Backup set	---	26-Sep-2017 22:09:17 v/d...
SSMCRestoreVSet0002...	Backup set	---	09-Oct-2017 02:09:55 v/d...
SSMCRestoreVSet0002...	Backup set	---	11-Oct-2017 02:02:58 v/d...
SSMCRestoreVSet0002...	Backup set	---	23-Sep-2017 01:04:39 v/d...
SSMCRestoreVSet0002...	Backup set	---	02-Oct-2017 23:01:52 v/d...
SSMCRestoreVSet0002...	Backup set	---	22-Sep-2017 19:37:29 v/d...
SSMCRestoreVSet0002...	Backup set	---	12-Oct-2017 02:10:54 v/d...

Selected Restore Point Details (Right Pane):

Backup:2017-08-19 00:05:08 Overview

General

- Name: Backup:2017-08-19 00:05:08
- Description: ---
- Creation time: 19-Aug-2017 00:20:04 v/dock IST
- Type: Backup set
- Backup store name: varaha_store1

Health

- Status: Normal
- Status description: Normal

Policy

- Protection: ps2-36-backup
- Incremental: true

Volumes

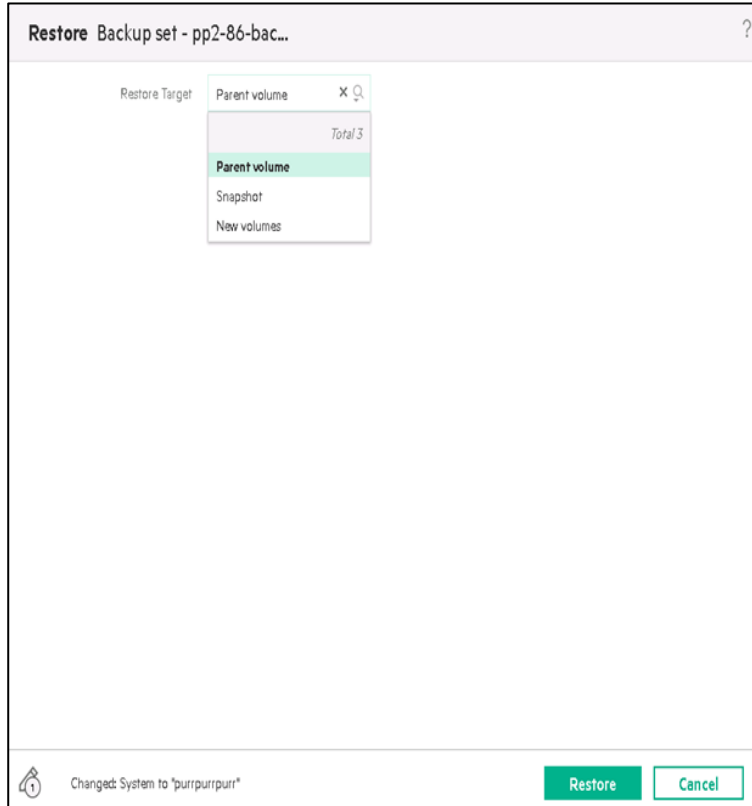
- Count: 2 volumes

Actions

- Attach
- Detach
- Restore
- Delete

- ▶ Restore points Overview shows all restore points.
- ▶ The Right Hand Side pane will show detailed information about that Restore Point
- ▶ From this page, the user can set an action for a Restore Point
 - ▶ Attach
 - ▶ Detach
 - ▶ Restore
 - ▶ Delete

Restore Point – Restore Action



- ▶ A Restore Point can be selected and restored via **Action- Restore**
- ▶ The Restore dialog shows three options:
 - ▶ To the parent volume
 - ▶ Snapshot
 - ▶ New Volume

Restore Points – Attach and Detach Action

Attach SSMCRecoverySet60002AC000000000020030110000B69ESnapshotSet-2017-09-30-22-00

Host

Name 🔍

Volume(s)

Name	Size	WWN
VVR1	18253611008	127606754668362479907820269569554056382

- ▶ A Restore Point can be attached to a Host
- ▶ If you provide the host name the object will be attached to the host
- ▶ Now you verify the content of the Restore Point through the host
- ▶ Click **Actions-Detach** to detach the host again

Restore Points – Delete Action

Delete SSMCRecoverySet60002AC000000000020030110000B69ESnapshotSet-... ?

Below restore point will be deleted and all data on the restore point will be lost. Restore point attached to a host will also be deleted. Do you want to continue?

Name	Type
SSMCRecoverySet60002AC000000000020030110000B69ESnapshotSet-2017-09-30-22-00-00	Snapshot set

- ▶ Select a Restore Point and click **Action-Delete** to delete a Restore Point
- ▶ Restore Point attached to a host will also be deleted





Priority Optimization

Priority Optimization

- ▶ Consolidation of storage systems reduces complexity of data storage and delivers efficiency in management, occupied floor space, and energy consumption. However, the consolidation of many disjoint workloads into a single storage system also results in contention for shared system resources on the system.
 - ▶ Examples of such shared resources include:
 - ▶ Front-end host Fiber Channel (FC), iSCSI and FCoE adapters
 - ▶ Back-end FC or SAS disk connections
 - ▶ Physical disks
 - ▶ Data and control cache
 - ▶ ASICs, CPUs, and backplane interconnections
-



PO

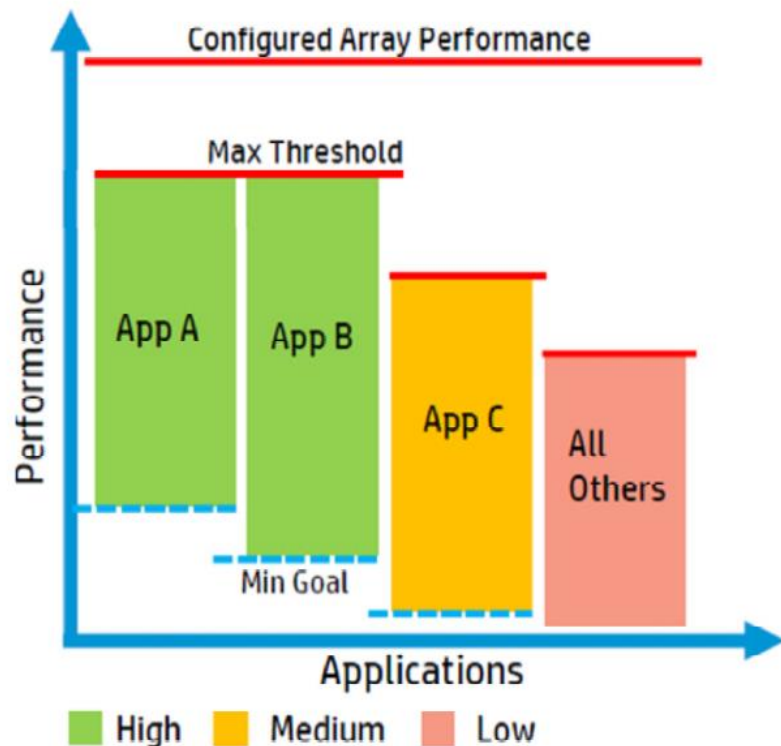
- ▶ Data packets arriving at the front-end FC HBA adapters are handled on a first-come, first-serve basis.
- ▶ However, it can lead to unequal and inconsistent throughput for multiple concurrent workloads.
- ▶ HPE 3PAR's Priority Optimization software manages and distributes the I/O capacity of an HPE 3PAR StoreServ Storage system across multiple workloads.
- ▶ The tool enables the co-location of the data of workloads of different types, such as sequential, random, online transaction processing (OLTP), with different I/O packet sizes on a single storage system while achieving adequate and stable performance in a multi-tenant environment.



Quality of Service (QoS) introduction

If all resources were unlimited from a performance perspective, there would be no need for QoS.
However:

- Storage resources are limited and expensive, and will have periods of contention that might disrupt business operations unpredictably
- Without storage QoS, hot or active data will use more storage performance resources, which can starve other storage resources
- Higher priority applications *could* fall below a desired **minimum** (if PO algorithm starts to throttle workloads) and lower priority applications could monopolize system resources (exceeding a desired **maximum**)



Why Quality of Service? (1 of 2)

Noisy neighbors disrupt cloud and enterprise storage delivery

When an application or VM spikes up its performance utilization and leaves less controller caching and fewer flash resources available for other applications, customers feel the adverse affects of *noisy neighbors*.

This limitation is a major reason why it is so difficult for cloud storage providers to create consistent, multitenant environments using traditional storage systems that lack Storage QoS

Throwing spindles or controllers to fix a performance problem is not always the best answer

Priority Optimization can be used to make sure the right customers and applications fully use the available resources while controlling less important applications



Why Quality of Service? (2 of 2)

HP 3PAR PO use cases:

- Prioritize mission and business critical workloads over testing and development
- Accommodate bursts in workloads
- Protection from rogue processes
- Manage application performance expectation levels
- Manage periodic variations in workloads
- Improving multi-tenant and cloud experience
- Managing Remote Copy replication traffic and Peer Motion migration traffic



How it works overview

Priority Optimization control is implemented between the target driver and VV layer

When receiving an I/O (read or write), QoS takes one of **three** actions:

- Pass the I/O request to the VV layer of the HP 3PAR StoreServ OS—no special delay
- Delay the I/O request by placing it in a private QoS queue that is processed periodically
 - When one or more of its applicable QoS rules have been breached, it will be delayed by putting it in a QoS queue until all of the QoS conditions are satisfied
 - This prevents hosts that over-issue I/Os from over-using their I/O quota and adversely affecting the performance of the system
- Return a SCSI Queue Full (QFULL) message to the server that sent the request
 - When too many I/Os are delayed, incoming I/Os are rejected using the queue full (SCSI QFULL) response sent to the host
 - This prevents delayed I/Os from holding all system resources such as host, HBA, and VV layer buffers and queues



Adaptive Flash Cache (AFC)

- ▶ HPE 3PAR Adaptive Flash Cache (AFC) is a built-in array functionality of the HPE 3PAR StoreServ that does this by using capacity on solid-state drives (SSDs) (flash) to act as Level-2 read cache holding random read data that has been removed from DRAM read cache.



Definition

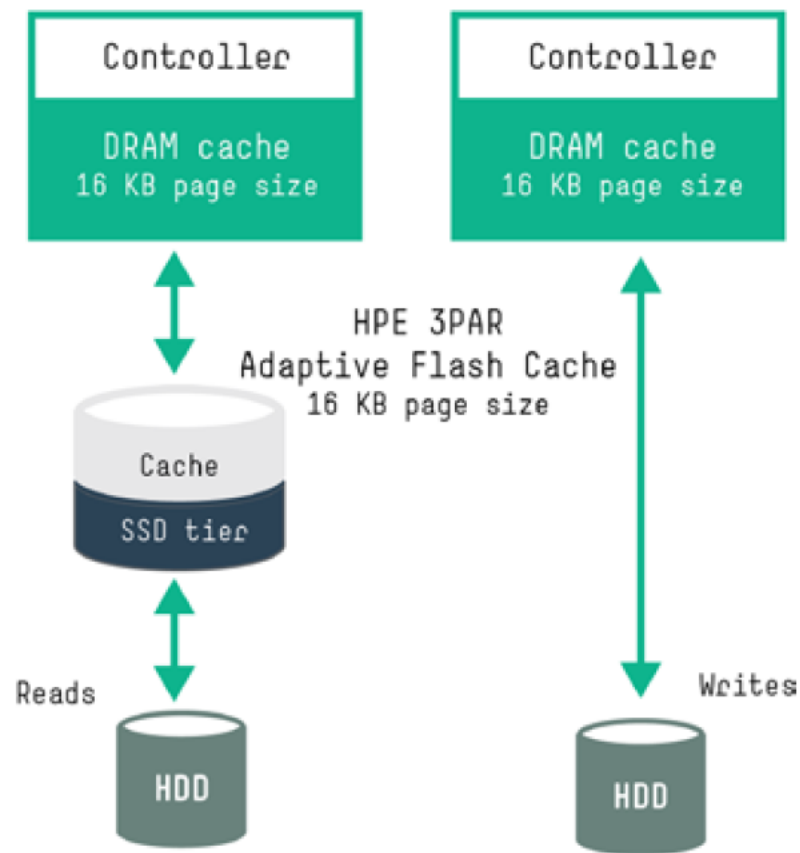
- ▶ **AFC**—Adaptive Flash Cache. HPE 3PAR implementation that uses flash (SSD) storage as a Level-2 read cache on an HPE 3PAR StoreServ array.
- ▶ **CMP**—Cache Memory Page. This refers to a 16 KiB page of physical DRAM cache memory that contains data for either a read IO or a write IO.
- ▶ **DRAM**—Dynamic random-access memory.
- ▶ **FMP**—Flash Memory Page. This refers to a 16 KiB page of physical flash in the Flash Cache.
- ▶ **SSD**—Solid-state drive. This “disk drive” uses non-volatile flash memory rather than a rotating platter to store blocks of data.



Without HPE 3PAR Adaptive Flash Cache



With HPE 3PAR Adaptive Flash Cache



```
3par-lab cli% showflashcache
Flash Cache is not present
3par-lab cli%
3par-lab cli% createflashcache 65536
3par-lab cli%
3par-lab cli% showflashcache
                        - (MB) -
Node Mode  State      Size Used%
   0 SSD   normal    32768    0
   1 SSD   normal    32768    0
-----
   2 total                65536
3par-lab cli%
3par-lab cli% █
```

3par-lab [cli% showld

Id	Name	RAID	--Detailed_State--	Own	SizeMB	UsedMB	Use	Lgct	LgId	WT
u	MapV									
2	.srdata.usr.0	1	normal	0/1	14336	14336	V	0	---	
N	Y									
3	.srdata.usr.1	1	normal	0/1	16384	16384	V	0	---	
N	Y									
4	.srdata.usr.2	1	preserved,checking	1/0	14336	14336	V	0	---	
N	Y									
5	.srdata.usr.3	1	preserved,checking	1/0	16384	16384	V	0	---	
N	Y									
0	admin.usr.0	1	preserved,checking	0/1	5120	5120	V	0	---	
N	Y									
1	admin.usr.1	1	normal	1/0	5120	5120	V	0	---	
N	Y									
11	fcacheld.0	0	normal	0/-	32768	0	FLC	0	---	
Y	N									
12	fcacheld.1	0	normal	1/-	32768	0	FLC	0	---	
Y	N									
6	log0.0	1	normal	0/-	20480	0	log	0	---	
Y	N									
7	log1.0	1	normal	1/-	20480	0	log	0	---	



LAB



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Create Policy

General ▾



Policies 0

Name ▾

+ Create policy

Name ▲ System

No policies

General

Policy type ☐ Virtual volume ☒ Snapshot

Snapshot policy name

sp1

Name must be unique. Name may be 1 to 31 alphanumeric characters including hyphen, period, and underscore but cannot begin with a hyphen.

Snapshot name pattern

Parent Volume & Time Stamp

Snapshot name

VolumeName.170605141808

Snapshot schedules

0 Schedules

Schedule ▲ Details
Name

Retention
Time

Expiration
Time

No schedules added.

Add schedule

Additional Settings

Create

Create +

Cancel

Create Snapshot Schedule



Schedule name

sch1

Schedule pattern

Hourly ▾

Starting at

0

minutes after the hour

Alert

Enabled ☒

Retention time

Enabled ☒

7

days ▾

The retention time cannot be removed or reduced once it is set. This volume cannot be removed until its retention time expires.

Expiration time

☐ Disabled

Create Policy

General ▾

General

Policy type ☐ Virtual volume ☒ Snapshot

Snapshot policy name

sp1

Snapshot name pattern

Parent Volume & Time Stamp ▾



Snapshot name VolumeName.170605141808

Snapshot schedules 1 Schedule

Schedule Name	Details	Retention Time	Expiration Time	
sch1	0 minutes after the hour	7 Days	—	×

Add schedule

Additional Settings



Policies 1

Name ▾

Type ▾

All ▾

+ Create policy

Name	System	Type
sp1	—	Snapshot

sp1

Overview ▾

General

Name

sp1

Type

Snapshot

Comments

—

Create Virtual Volume

General ▾



General

RAID 5 FC

☐ Advanced options

Size

1

GiB ▾

App Volume set

Search



Protection And Additional Settings

► Volume

▼ RMC Protection

Protection policy *no RMC instance configured. Click [here](#) to configure.*

▼ Snapshots

Snapshot policy

sp1



optional

► Remote Copy Group



Changed: Snapshot policy to "sp1"

Create

Create +

Cancel

Create Virtual Volume

General ▾



General

☐ Advanced options

Protection And Additional Settings

▼ Volume

Number of volumes

1

Volume set

Search



optional

Copy CPG

SSD_r5



optional

RAID 5 SSD

Comments

optional

▼ RMC Protection

Protection policy no RMC instance configured. Click [here](#) to configure.



Changed: Copy CPG to "SSD_r5"

Create

Create +

Cancel

Convert Virtual Volumes

General ▾



General

System dxc-3par ▾

Domain < None > ▾

Virtual Volumes

Conversion

Conversion type To fully provisioned ▾

Dedup —

Volume Name ▲	Volume Set	Provisioning	Dedup	Virtual Size (GiB)	Reserved Size (GiB)	User CPG	Copy CPG	
vvol2	—	Thin	No	1.00	1.13	FC_r5	SSD_r5	×

Add virtual volumes



Changed: Conversion type to "To fully provisioned"

Convert

Cancel

```
dx-c3par cli%
```

```
dx-c3par cli% createsv -ro snap1 vvol2
```

```
dx-c3par cli%
```

```
dx-c3par cli% showvv snap1
```

Id	Name	Prov	Type	CopyOf	BsId	Rd	-Detailed_State-	-Rsvd(MB) --	- (MB) -
4	snap1	snp	vcopy	vvol2	3	RO	normal	--	--

1 total								0	0

```
dx-c3par cli%
```

```
dx-c3par cli%
```

```
dx-c3par cli% createsv snap2 snap1
```

```
dx-c3par cli%
```

```
dx-c3par cli% showvv
```

```
      ---Rsvd(MB)---  -(MB)-
Id Name      Prov Type  CopyOf BsId Rd -----Detailed_State-----
----- Adm  Snp    Usr  VSize
  1 .srdata full base  ---      1 RW not_started,internal_consistency_error,snap
data_invalid  0    0 61440  61440
  0 admin    full base  ---      0 RW not_started,internal_consistency_error,snap
data_invalid  0    0 10240  10240
  2 vvol1    tpvv base  ---      2 RW not_started,internal_consistency_error,snap
data_invalid 128  512  512   1024
  3 vvol2    tpvv base  ---      3 RW normal
      256  512  512   1024
  4 snap1    snp  vcopy vvol2    3 RO normal
      --  --  --      1024
  5 snap2    snp  vcopy snap1     3 RW normal
      --  --  --      1024
-----
```

```
dxs-3par cli% createsv -ro snap3 snap2
```

```
dxs-3par cli%
```

```
dxs-3par cli% showvv vvol2
```

Id	Name	Prov	Type	CopyOf	BsId	Rd	-Detailed_State-	-Rsvd(MB)-	--	-(MB)-	
3	vvol2	tpvv	base	---	3	RW	normal	Adm	SnP	Usr	VSize
								256	512	512	1024

1	total							256	512	512	1024

```
dxs-3par cli% showvv snap*
```

Id	Name	Prov	Type	CopyOf	BsId	Rd	-Detailed_State-	-Rsvd(MB)-	--	-(MB)-	
4	snap1	snp	vcopy	vvol2	3	RO	normal	Adm	SnP	Usr	VSize
								--	--	--	1024
5	snap2	snp	vcopy	snap1	3	RW	normal	--	--	--	1024
6	snap3	snp	vcopy	snap2	3	RO	normal	--	--	--	1024

3	total							0	0	0	3072

dxc-3par cli%

dxc-3par cli% createvvcopy -p volc1 volclone1

Copy was started. child = volclone1, parent = volc1, task ID = 121

dxc-3par cli% showtask 121

No tasks.

dxc-3par cli%

dx-c3par cli% showvv

		---Rsvd(MB) ---			-(MB) -		
Id	Name	Prov	Type	CopyOf	BsId	Rd	-----Detailed_State----
		Adm	Snp	Usr	VSize		
1	.srdata	full	base	---	1	RW	not_started,internal_consistency_errc
	r,snapdata_invalid	0	0	61440	61440		
0	admin	full	base	---	0	RW	not_started,internal_consistency_errc
	r,snapdata_invalid	0	0	10240	10240		
9	volc1	cpvv	base	---	9	RW	normal
		256	256	512	512		
10	vvcp.9.8	snp	vcopy	volc1	9	RO	normal
		--	--	--	512		
8	volclone1	full	pcopy	vvcp.9.8	8	RW	copy_target
		0	0	2048	2048		
2	vvoll	tpvv	base	---	2	RW	not_started,internal_consistency_errc
	r,snapdata_invalid	128	512	512	1024		
3	vvoll2	tpvv	base	---	3	RW	normal
		256	512	512	1024		
4	snap1	snp	vcopy	vvoll2	3	RO	normal
		--	--	--	1024		
5	snap2	snp	vcopy	snap1	3	RW	normal
		--	--	--	1024		

dx-c3par cli% promotesv snap2

Task 122 has been started to promote virtual copy snap2

dx-c3par cli%

dx-c3par cli%

dx-c3par cli%

dx-c3par cli% showvv

		---Rsvd (MB) ---			- (MB) -			
Id	Name	Prov	Type	CopyOf	BsId	Rd	Detailed_State	
		Adm	Snp	Usr	VSize			
1	.srdata	full	base	---	1	RW	not_started,internal_consistency_error,snapdata_invalid	
0	admin	full	base	---	0	RW	not_started,internal_consistency_error,snapdata_invalid	
9	volc1	cpvv	base	---	9	RW	normal	
		256	256	512	512			
10	vvcp.9.8	snp	vcopy	volc1	9	RO	normal	
		--	--	--	512			
8	volclone1	full	pcopy	vvcp.9.8	8	RW	copy_target	
		0	0	2048	2048			



GENERAL

Dashboard

Activity

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Restore Points 3

Status

All

Name

Type

All

All systems

	Name	Type	System	Creation Time
●	snap3	Snapshot	dxc-3par	Jul 6, 2020 11:04:56 AM
●	snap2	Snapshot	dxc-3par	Jul 6, 2020 11:02:55 AM
●	snap1	Snapshot	dxc-3par	Jul 6, 2020 11:00:58 AM

✓ snap3

Overview

General

Name	snap3
ID	6
System	dxc-3par
WWN	60002AC00000000000000000600...
Virtual size	1.00 GiB
Type	Snapshot
Parent	snap2

Health

State	● Normal
State description	Normal

Copies

Clones	0
Snapshots	0

Actions

Attach

Detach

Restore

Delete

```
dx-3par cli% setqos -io 300 vvset:dbl
```

```
dx-3par cli%
```

```
dx-3par cli% showqos
```

Id	Type	Name	QoS	Priority	-I/O_per_second-		-KBytes_per_sec-		LatencyGoal
					Min	Max	Min	Max	
0	vvset	dbl	on	normal	300	300	-	-	-

```
1 total
```

```
dx-3par cli%
```

```
dx-3par cli%
```

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```
dxs-3par cli%
dxs-3par cli% setqos -clear vvset:*
dxs-3par cli%
dxs-3par cli% showqos
No QoS rules listed
dxs-3par cli%
dxs-3par cli%
```



Create Priority Optimization Policy

General ▾



General

System dxc-3par ▾

Target type App Volume Set ▾

Target name db1 × 🔍

State Enabled

Optimization Settings

Priority High ▾

IOPS maximum limit 1000 IO/s

IOPS minimum goal 500 IO/s

Bandwidth maximum limit 4096 KB/s



Changed: Bandwidth minimum goal to "3000"

Create

Create +

Cancel

db1

Overview ▾



- Create Priority Optimization Policy (db1) Completed

General

Target name	db1
System	dxc-3par
Target type	App Volume Set
App volume set	db1
Target name on system	db1
State	Enabled
Priority	High
IOPS minimum goal	500 IO/s
IOPS maximum limit	1,000 IO/s
Bandwidth minimum goal	3,000 KB/s
Bandwidth maximum limit	4,096 KB/s
Latency goal	2.00 ms